SVM - CYP regression results

Main result

We used chemprop (not SVM's!) as the foundation for model development.

The different model types and their training procedures are described below.

Model ensemble A:

We extended chemprop with a pretraining and fine-tuning pipeline, a customized loss function, and additional descriptors.

We used a **linked loss** for 1A2 and 2D6 combining direct pIC50 supervision with a normalized log2FC screening term weighted by 0.30:

$$L = L_{\text{pIC50}} + 0.30\, L_{\text{log2FC, normalized}}$$

$$\log_2\text{FC (predicted)} = -\log_2(1 + C \times 10^{\text{pIC50 (predicted)}})$$

Here, C is the screening concentration in molar units. The conversion assumes a Hill slope of 1 and complete inhibition. Each loss term uses only compounds for which the corresponding measurement is available.

Selected descriptors added to model A:

We trained a separate ExtraTrees model per CYP using 5,939 candidate features, targeting direct pIC50 for 1A2, 2C9 and 3A4, and log2FC screen values for 2D6. We ranked features by the increase in mean absolute error (MAE) on the held-out validation set after shuffling each feature and selected the top 64 for each CYP. The combined feature sets for 1A2, 2C9 and 3A4 contained 154 distinct features, which were used for neural-model fine-tuning. For 2D6, we retained a separate set of 64 features.

Model A training proceeded as follows:

Models were initialized from the externally pretrained checkpoint, then finetuned using a shared 12-task model for 1A2, 2C9 and 3A4, with a separate CYP2D6-local model. Each endpoint subsequently underwent independent continuation using only its own CYP-labelled data, masking other CYP losses while keeping the transferred network trainable. Final training used the fully available training pool, fixed previously established durations, batch size 512 and 5 seeds; continuation used a constant learning rate of 3.79×10^{-5}.

CYP	Base training	CYP-specific continuation	Loss during continuation	Total epochs
1A2	144 epochs, joint model	27 epochs	Direct pIC50 plus independent screen/Emax auxiliary losses; the linked screen loss was disabled	**171**
2C9	144 epochs, joint model	11 epochs	Direct pIC50 plus independent CYP2C9 auxiliary losses	**155**
2D6	50 epochs, separate local model	57 epochs	Direct pIC50 plus 0.30 × normalized linked log2FC loss	**107**
3A4	144 epochs, joint model	33 epochs	Direct pIC50 plus independent CYP3A4 auxiliary losses	**177**

Model ensemble B

We fine-tuned the previous graph models with two additional loss terms, tackling both assay uncertainty and potency differences between close analogs

Assay uncertainty

For each molecule, the model predicts pIC50 p and an intrinsic standard deviation σ. During training, we combine σ with the assay's reported standard deviation s:

$$v = \sigma^2 + s^2$$

$$L_\text{Gaussian} = \tfrac{1}{2}[\log(v) + (p - y)^2/v]$$

Here, y is the measured pIC50. The two terms have different jobs:

- Error divided by variance: reduces the pressure to fit noisy measurements precisely.
- Log-variance penalty: discourages the model from avoiding errors simply by predicting very large uncertainty.

Analog potency difference

For a close-analog pair (mol A and mol B) with confidence intervals [l_A, u_A] and [l_B, u_B], the predicted potency difference should fall within:

$$P_A - p_B \in [l_A - u_B, u_A - l_B]$$

Training uses AdamW, batch size 32, encoder/head learning rates of $10^{-5}/10^{-4}$, and at most 20 epochs with patience five. The predictions from the fine-tuned models were added as a small correction to earlier results.

Model ensemble C

We fine-tuned the previous graph models with regularization for CYP1A2 and an ordinal head for CYP2D6.

Regularization

1. **Representation regularization.** We passed the same molecule through the model being fine-tuned and a frozen pretrained encoder. We penalized differences between their L2-normalized molecular vectors. This encourages the updated encoder to preserve useful molecular representations while learning the CYP task.
2. **Weight regularization.** We penalized changes in the graph encoder's parameters relative to their pretrained values. This allows adaptation but makes large departures from the pretrained parameters costly.

Ordinal head

Instead of directly outputting one pIC50 number, it predicted probabilities across a grid of possible pIC50 values. The supplied confidence bounds defined the training target: half the target mass was placed near the lower bound and half near the upper bound. We trained the graph encoder and this head together, then converted the predicted distribution into a single potency using its median to target the expected outside-interval absolute error.

we fine-tuned the graph encoder and prediction heads using AdamW, with learning rates of 10^{-5} for the encoder and 10^{-4} for the heads, weight decay of 10^{-4}, and batch size 32. The ordinal head used a 128-unit hidden layer and a 0.05-pIC50 grid. We allowed up to 12 epochs, with early-stopping patience of 3 after at least eight epochs. The predictions from the fine-tuned models were combined with earlier results.

Final results were blends of models as shown in the section below.

How the predictions were combined

Ensemble A is the reference combination. Ensemble B adds predictions informed by molecular similarity and assay uncertainty. Ensemble C adds complementary predictions from other model variants. Each ensemble already combines several models; the percentages below describe prediction-file contributions.

Enzyme	Ensemble A	Ensemble B	Ensemble C
CYP1A2	75%	0%	25%
CYP2C9	50%	2.90%	47.10%
CYP2D6	65%	26.11%	8.89%
CYP3A4	75%	25%	0%

CYP2C9 uses the original prediction values. For the other enzymes, predictions are first aligned to Ensemble A's numerical scale while retaining their order. Changes from Ensemble A are capped at ±0.25 pIC50. The final candidate copies the accepted enzyme columns exactly and trains no additional model.

The best results per CYP were chosen on the intermediate leaderboard – based on the ST-RAE, as follows:

Best recorded result for each enzyme

Enzyme	ST-RAE	MAE	R^2	Accepted source
CYP1A2	0.41	0.64	0.66	31 October
CYP2C9	0.32	0.39	0.74	2 November
CYP2D6	0.47	0.86	0.43	2 November
CYP3A4	0.38	0.45	0.75	31 October

What improved performance

The strongest recorded gains came from retaining successful predictions for each enzyme, adding stronger pretrained predictions, and keeping blend corrections small. The later CYP2C9 and CYP2D6 changes improved their public leaderboard scores. Changes that worsened CYP1A2 or CYP3A4 were left out of the final candidate.

More complex models and additional descriptors did not consistently help. Some experiments improved average-error measures while worsening ST-RAE. Model selection therefore followed the challenge metric rather than assuming that greater complexity meant better predictions. Finally most performance was gained from focusing on the ST-RAE not on the error / correlation. We also observed that validation performance for 2D6 and 1A2 was not necessarily correlated with leaderboard performance.